

Lesson 5-4 · Period 2

Radical Equations With Rational Exponents + DOK-3 Half-Life
Model

Klmsara-adapted · Student-centered · No Blooket

Do Now — Solve the radical equation

Algebra 2 · Lesson 5-4 · Period 2

DOK 1

5 min

bridge from P1

Practice 5-4-savvas-q21 · Cube-root equation Solve each radical equation.

$$\sqrt[3]{x} + 8 = 13$$

Use the sentence frame: “I isolate the radical, then raise both sides to the power ____.”

Launch — Example 4 (Rational-Exponent Equations)

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DOK 2

12 min

teacher models

Rational-Exponent Rules (keep on your page)

1. Rewrite $x^{m/n}$ as $(\sqrt[n]{x})^m$ — bridge to radicals you know.
2. To solve $(\text{expr})^{m/n} = k$, raise both sides to reciprocal power n/m .
3. Simplify, then **CHECK** — extraneous solutions arise when denominator is EVEN.
4. For decay: $y = a \cdot (0.5)^{(t/h)}$ — h is the half-life (denominator of exponent).

EXTRANEIOUS SOLUTIONS: If n is EVEN, always substitute back into the original.

Explore — TPS + DOK-3 Half-Life Model

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DOK 2-3

33 min

student work

5 items in order. Plan ~12 min for Practice #41 (Half-Life Model).

Try It 4 · Rational-exponent equations Solve: $(x^2 - 3x - 6)^{3/2} - 14 = -6$ and $(x - 8)^{3/2} = (10 - x)^3$.

Practice #12 · Solve with rational exponent $(3x + 2)^{2/3} = 4$ — does the denominator hint at extraneous solutions?

Practice #33 · Rational-exponent equation $0.5(x^2 + 5x + 136)^{2/3} = 50$ — CHECK both solutions.

DOK 2

5 min

DOK-3 reveal + P3 bridge

Sentence frame · Half-Life Model “This equation has exponent ____/____, so I raise both sides to the power ____/____. After simplifying I get _____. In the half-life formula, $a = \text{---}$ (start), $h = \text{---}$ (half-life period), $t = \text{---}$ (elapsed time), giving $y \approx \text{---}$.”

Bridge to Period 3 Next period: apply rational-exponent skills to **assessment-critical** items.

Bring your check-for-extraneous habit.

DOK 1-2

3 min

not graded

Complete on your exit ticket

1. To solve $(\text{expr})^{m/n} = k$ I raise both sides to power _____.
2. Half-life tells me the _____ of the exponent.
3. After 16 hours, 50 mL of soft drink decays to _____ mL.
4. One biggest thing I learned today: _____.